AI WAN Key Technologies for Distributed Inference

zhangka@huawei.com
gengxuesong@huawei.com

Security Level:

HUAWEI

Contents

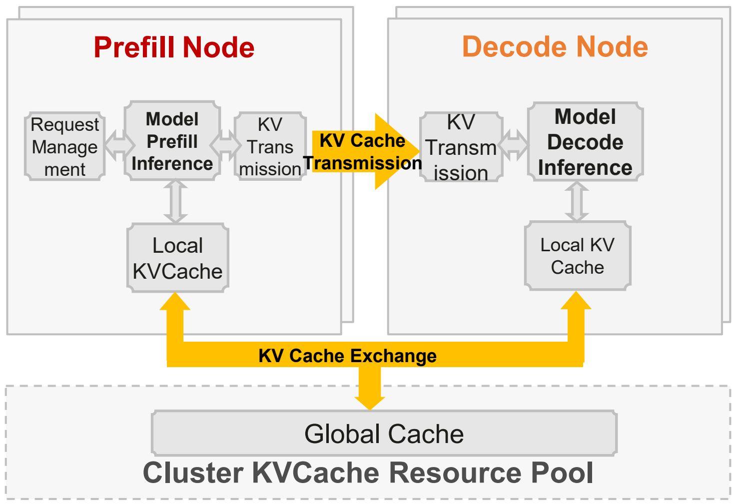
HUAWEI

AI Service Evolution and Emerging Network Requirements

Phase 1

Intelligent Computing Cluster

Inside the AI cluster

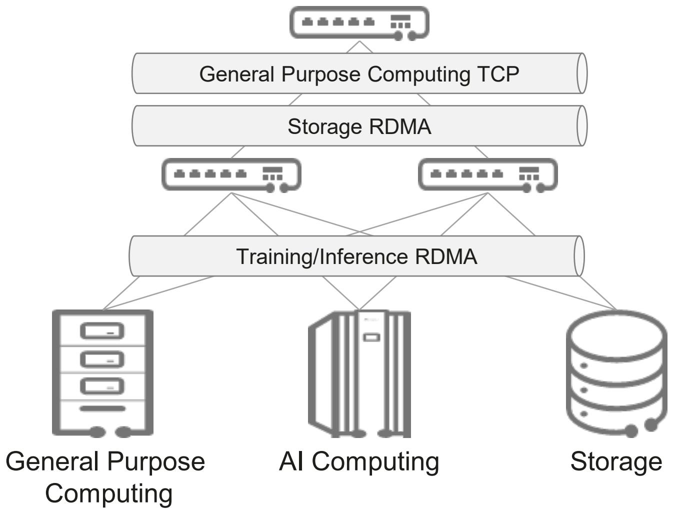


Inference technology has evolved from single-machine inference to cluster-based inference. Through multi-machine collaboration, improving the performance and cost-effectiveness of inference.

Phase 2

AIDC Integration of General Purpose, AI Computing and Storage

Inside the Data Center

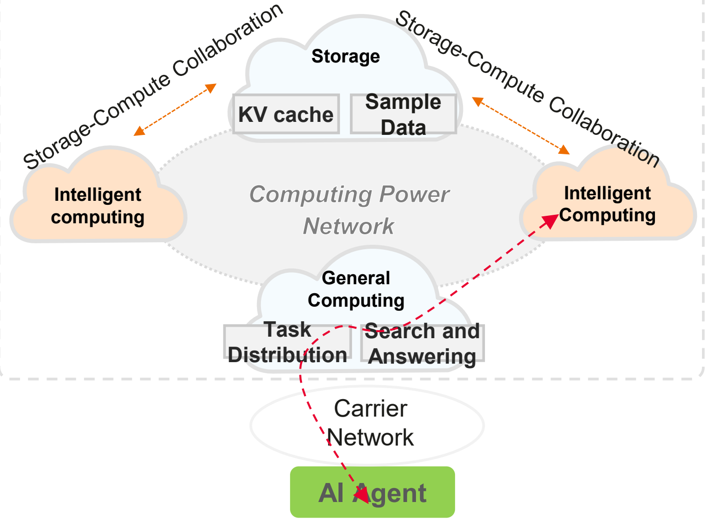


The data center is evolving toward a convergence of general purpose computing, intelligence, and storage, with a high-speed interconnect bus as core, significantly enhancing the performance and cost-effectiveness of inference.

Phase 3

Cross-DC Collaboration of General Purpose, AI Computing and Storage

Cross-domain integrated computing power supply

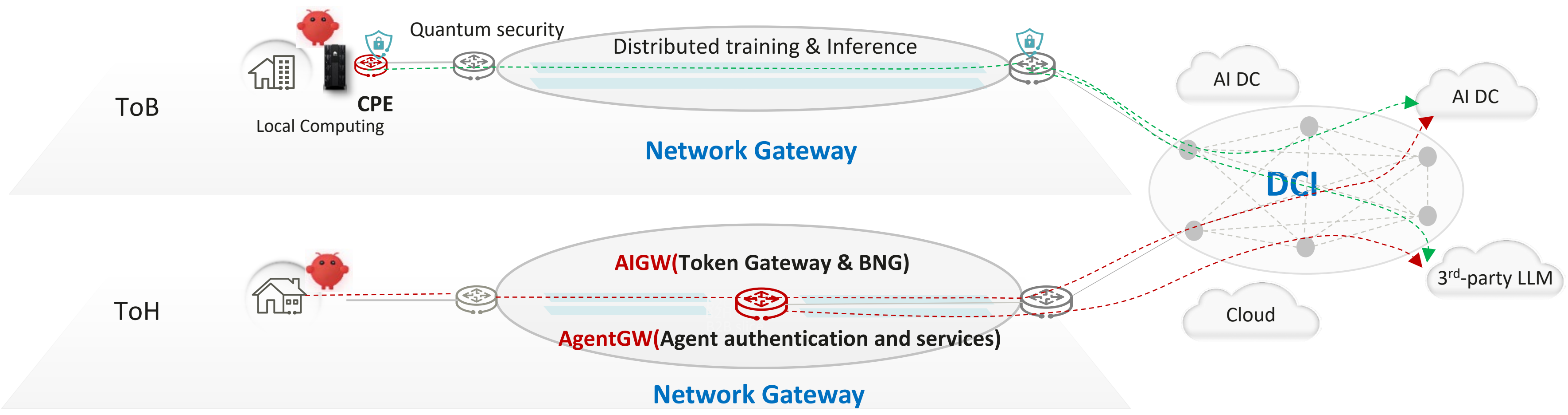


The interconnection of computing power across regions and across diverse forms of computing power “General purpose-intelligence-storage“, breaks down the spatial and structural barriers of computing power resources, enabling efficient collaboration of computing power.

HUAWEI

Network Requirements for Distributed AI

New Service Enabled by DCA and DCI Networks



Elastic & lossless AI Inference

- Hierarchical deployment of AI model for distribute inference
- Subscriber Precise flow control, computing Efficiency 95%+

Optimal experience & cost-efficient token service

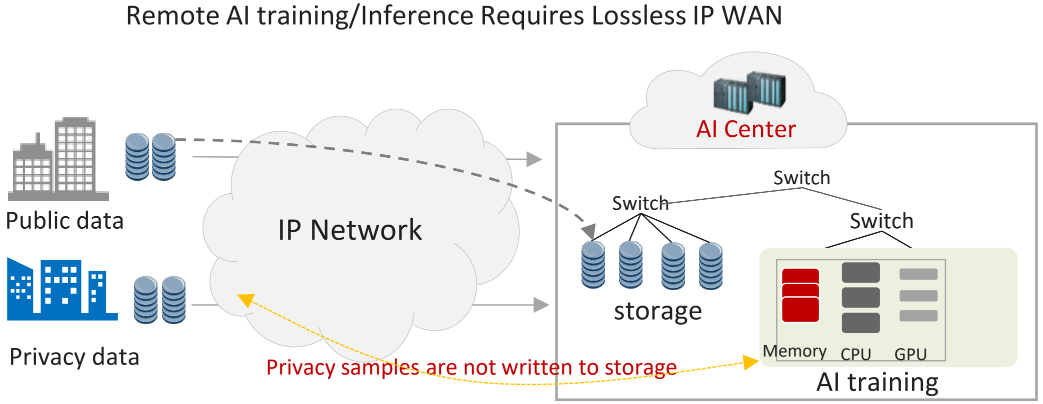
- Seamless token access with unified API key and converged HBB & token billing
- Integrated computing & network scheduling with semantic-aware model routing
- Differentiated SLA guarantee for token interaction

Agent Communication and Authentication

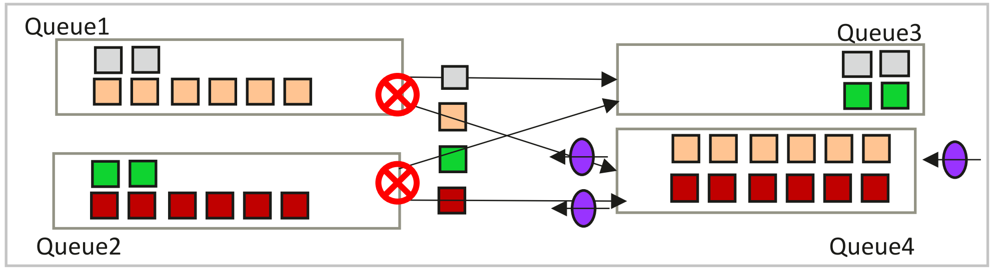
- Cross-platform heterogeneous interconnection of multiple agents, secure Low-Latency Communication
- Agent interaction security access control and management

Lossless Distributed inference: SPFC achieve precise flow control, 95% high throughput

Remote AI training/Inference, multi-tenant throughput decreases

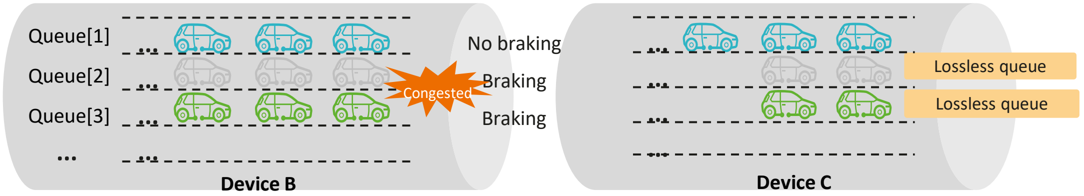


- Traditional solutions use PFC to achieve RDMA lossless transmission, but suffer from Head-of-Line (HOL) blocking.
- When congestion occurs in one flow, all tenants sharing the same port and priority are impacted.



SPFC achieve precise flow control, 95% high throughput

- **Precise flow control**: Uses **tenant-level elastic slices** to isolate services between tenants sharing the same port. Uses paths and queues to implement tenant-level flow control.
- **Long-distance lossless transmission**: Uses automatic backpressure and hop-by-hop rate reduction to absorb burst traffic through network-level buffer.



- Using **ICMPv6 header extension** to carry backpressure information

IPv6 header	SRH	ICMPv6 Header	IPv6 Payload

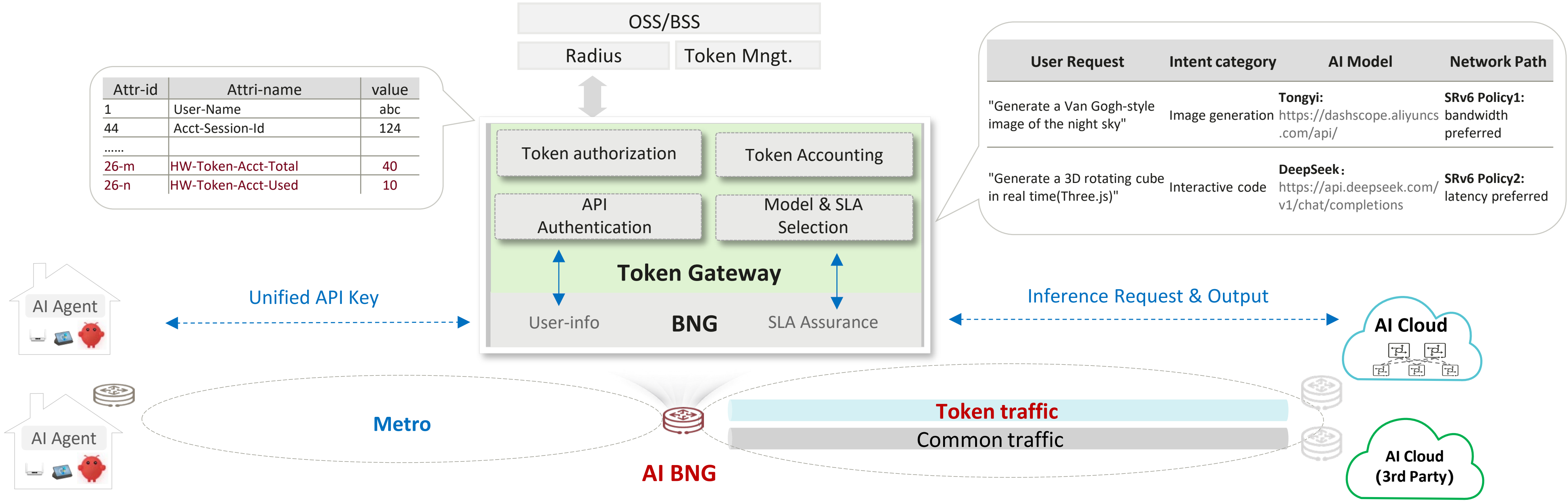
0 7 15 23 31

Type	Code	Checksum
Flag	Priority	reserve
Flow-Control-Argument [0]		Flow-Control-Argument [1]
Flow-Control-Argument [2]		Flow-Control-Argument [3]
Flow-Control-Argument [4]		Flow-Control-Argument [5]
Flow-Control-Argument [6]		Flow-Control-Argument [7]
Backpressure Bandwidth		
Slice ID		

https://datatracker.ietf.org/doc/draft-liu-rtgwg-srv6-cc/: Congestion Control Based on SRv6 Path

AI Gateway: One-stop Service Based On BNG Integrating Model And Network Capability

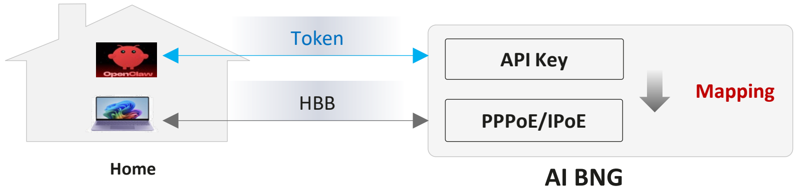
Integrated Token Service For Home Broadband Users



Attr-id	Attri-name	value
1	User-Name	abc
44	Acct-Session-Id	124
……		
26-m	HW-Token-Acct-Total	40
26-n	HW-Token-Acct-Used	10

User Request	Intent category	AI Model	Network Path
"Generate a Van Gogh-style image of the night sky"	Image generation	**Tongyi:** https://dashscope.aliyuncs.com/api/	**SRv6 Policy1:** bandwidth preferred
"Generate a 3D rotating cube in real time(Three.js)"	Interactive code	**DeepSeek:** https://api.deepseek.com/v1/chat/completions	**SRv6 Policy2:** latency preferred

Unified Authentication & Accounting

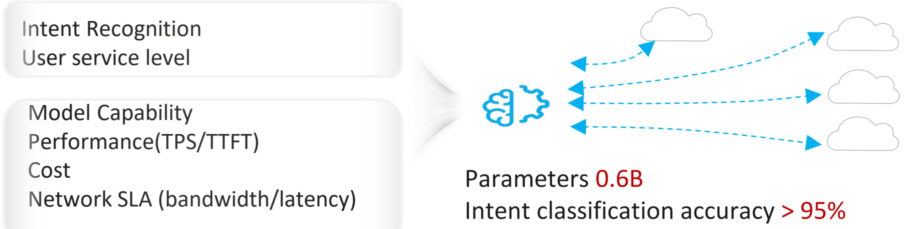


Token auto-authentication with API Key & PPPoE/IPoE mapping

Seamless accesss & Simplified integration

Integrated Computing & Network Scheduling

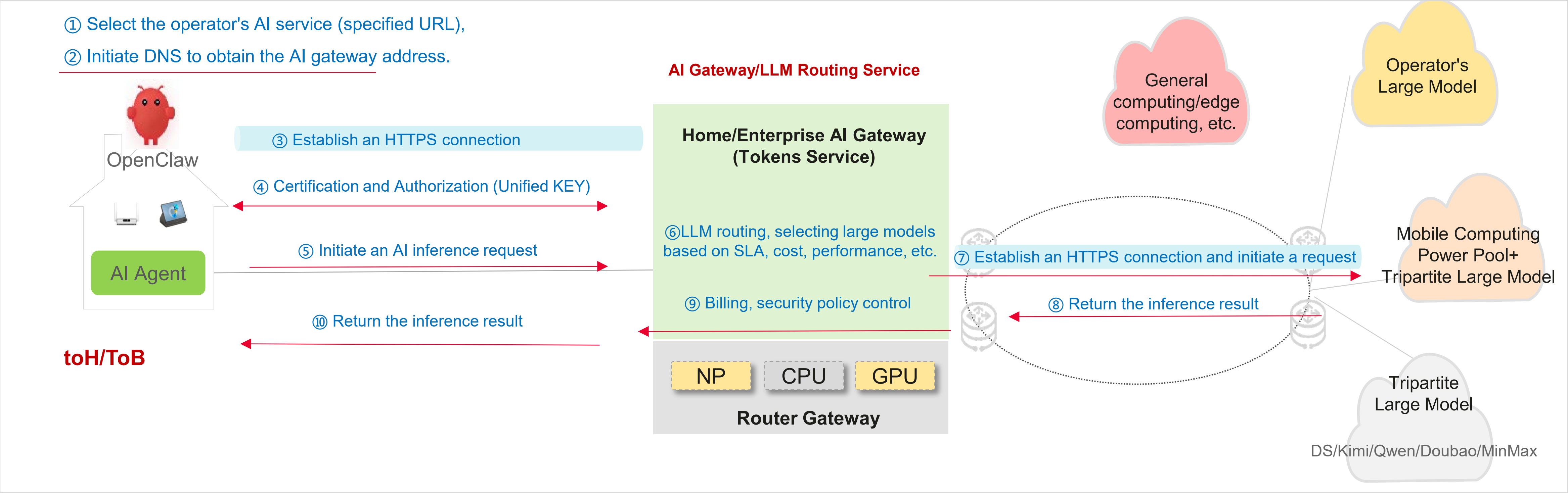
Semantic-aware model routing



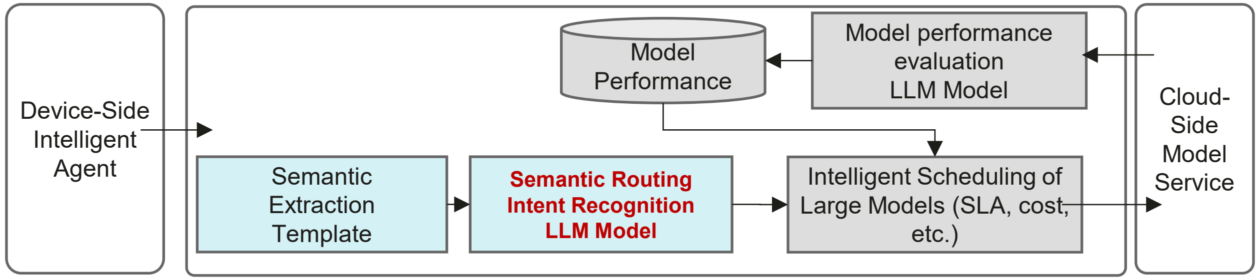
Optimal TCO & Experience assurance

HUAWEI

AI Gateway key technologies: Integration of router gateways with AI gateways, enabling intelligent token perception and the integration of computing and network scheduling



AI Gateway: Based on a semantic-aware large model for inference tasks, it implements LLM routing and optimizes token costs by 20% → 40%+



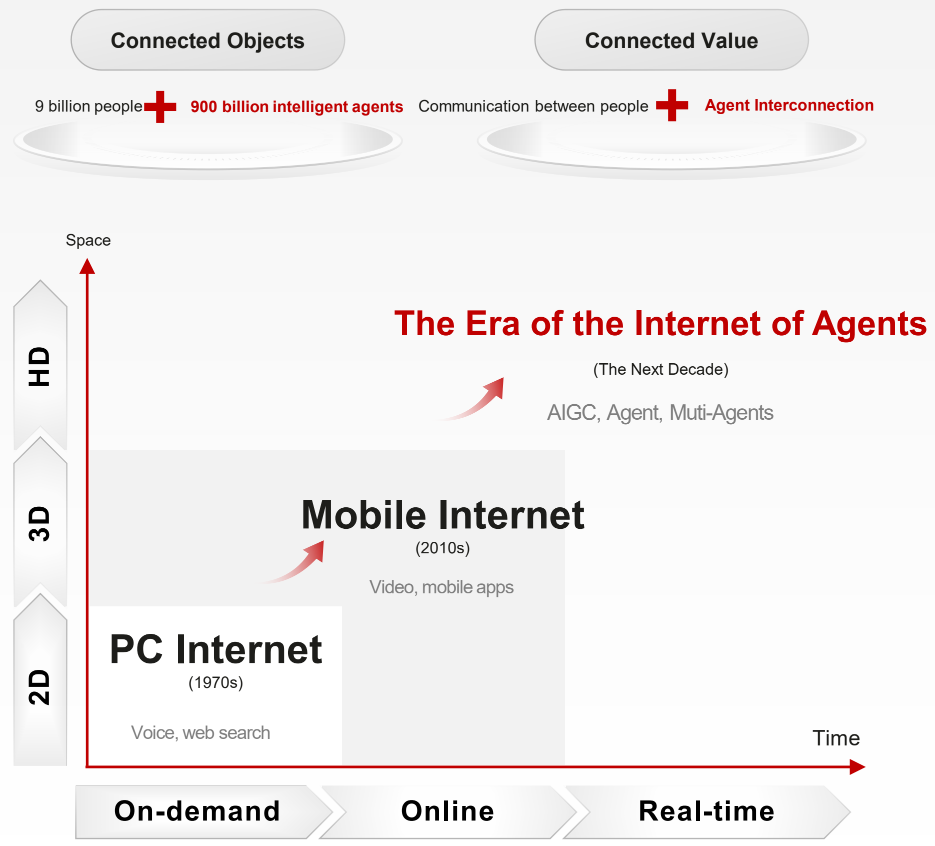
Concurrency requirement calculation: 30,000 users, 2 agents per user, 10% concurrency, task every 5 seconds

Business Requirement	Performance Requirement	Model Type	Parameters	Computing Power Demand
Semantic Awareness Routing LLM Model	1200/s <10ms latency 1 ms in some scenarios	Decoder-only Transformer	0.6B	140 TFlops

HUAWEI

Agent Gateway: IOA creates new infrastructure requirements, positioning the Agent Gateway as the core entry point

An intelligent agent internet enables ultra-dimensional interconnection of all things and real-time interaction between the virtual and physical worlds.



The intelligent agent internet requires new intelligent infrastructure.

Network Challenges

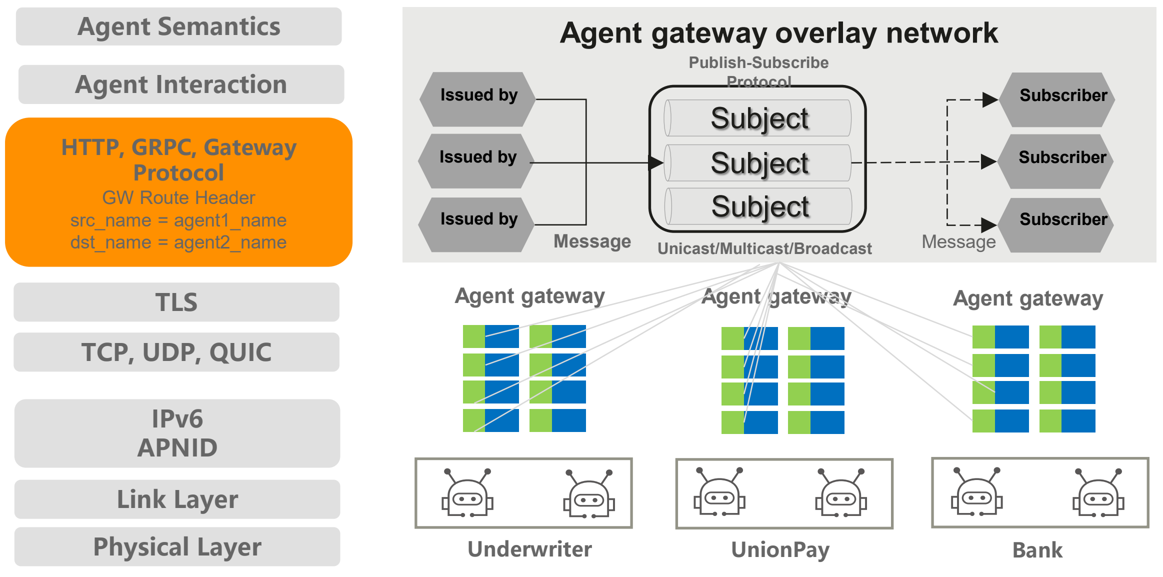
Communication mode	Communication protocol	Communication quality
High concurrency and massive connections	High-performance semantic addressing	Cross-domain machine-machine interaction quality
Traditional application point-to-point → Mesh between agents	IP routing addressing → Agent semantic addressing	For every 50ms increase in communication time,
The number of sessions increases exponentially (n x m).	**Addressing speed increased by 1000 times**	**The efficiency of multi-agent systems decreases by an average of 10%**

Characteristics of the Internet of Intelligence

Ultra-band	Intelligence	Determinism	Green	Safety
Next 10 years 100-fold traffic growth	Intelligent addressing Resource adaptation	Cross-layer collaboration End-to-end experience assurance	10x capacity increase Unchanged power consumption	Network authentication Security and intelligence board

Agent Gateway Key Technologies: Cross-platform heterogeneous interconnection and Agent interaction security control and management

Cross-platform heterogeneous interconnection of multiple agents



Values of Agent Gateway Communication Network

Heterogeneous Agent Interconnection

Interconnection of heterogeneous intelligent agents across organizations/platforms

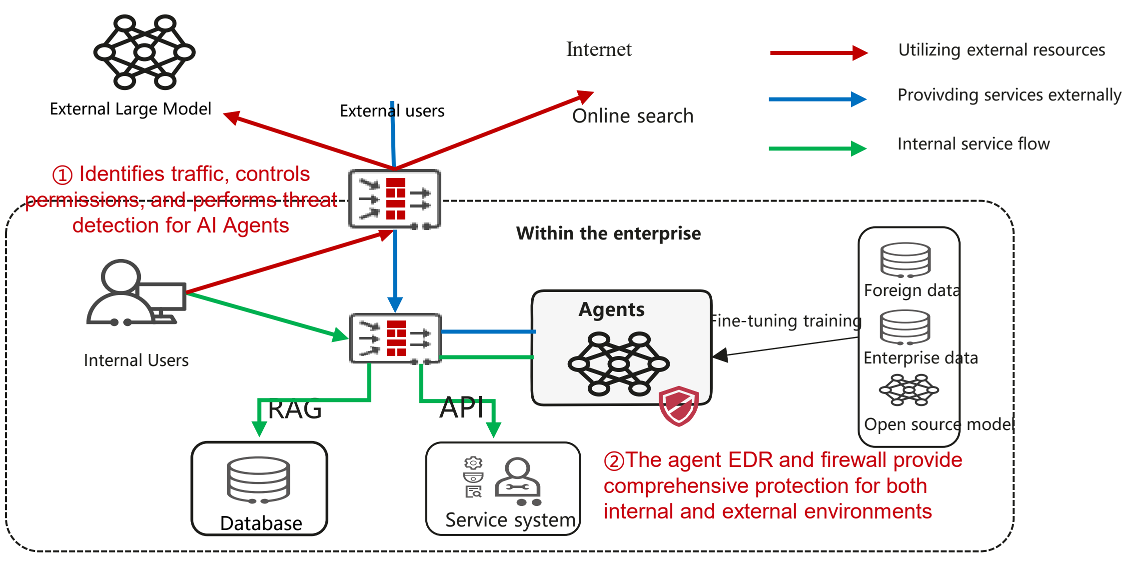
Reliability Assurance for Long-Task Sessions

Dynamic joining and exiting of intelligent agents, resumption from breakpoints, and uninterrupted task execution.

Secure Low-Latency Communication

Supports synchronous and asynchronous communication modes, as well as unicast and multicast message distribution.

Agent interaction security control and management



Value of Intelligent Security Management and Control

Intrusion Prevention

System intrusion prevention, boundary protection, prompt injection attack prevention

Permission Control

Agent identification and internal permission control, API security

Zero Data Leakage

Sensitive data identification, dynamic data classification and control

HUAWEI

Summary and Future Standardization Directions

- Distributed AI inference and agent collaboration introduce a new WAN networking scenario beyond traditional IP connectivity.
- This scenario requires a coordinated network architecture covering different scenarios and use cases.
- New protocol extensions are needed for example: AI service discovery and intelligent service routing.
- The scenario, architecture, and protocol extensions request a discussion home in IETF

Thank you.

把数字世界带入每个人、每个家庭、
每个组织，构建万物互联的智能世界。

Bring digital to every person, home and organization for a fully connected, intelligent world.

HUAWEI